

lecture 16.

Equivalence Relation.

- Reflexive
- Symmetric
- Transitive.

Ex2 :- $R = \{(a,b) \mid a-b \in \mathbb{Z}\} \quad A = \mathbb{R}.$

Reflexive :- $\forall a \in A \quad (a,a) \in R.$
 $\forall a \in \mathbb{Z} \quad a-a \in \mathbb{Z} \quad \checkmark$

Symmetric $\forall a,b \in A \quad \text{if } (a,b) \in R \rightarrow (b,a) \in R.$
 $\forall a,b \in \mathbb{Z} \quad a-b \in \mathbb{Z} \rightarrow b-a \in \mathbb{Z} \quad \checkmark$

Transitive $\forall a,b,c \in A \quad \text{if } (a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$
 $\forall a,b,c \in \mathbb{Z} \quad \text{if } a-b \in \mathbb{Z} \wedge b-c \in \mathbb{Z} \rightarrow a-c \in \mathbb{Z} \quad \checkmark$

It is Equivalence Relation.

Ex3/494 $R = \{(a,b) \mid a \equiv b \pmod{m}\} \quad A = \mathbb{Z}.$

Reflexive :- $\forall a \in A \quad (a,a) \in R.$
 $\forall a \in \mathbb{Z} \quad a \equiv a \pmod{m}.$

Symmetric $\forall a,b \in A \quad \text{if } (a,b) \in R \rightarrow (b,a) \in R.$
 $\forall a,b \in \mathbb{Z} \quad \text{if } a \equiv b \pmod{m} \rightarrow b \equiv a \pmod{m} \quad \checkmark$

Transitive $\forall a,b,c \in A \quad \text{if } (a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$
 $\forall a,b,c \in \mathbb{Z} \quad \text{if } a \equiv b \pmod{m} \wedge b \equiv c \pmod{m} \rightarrow a \equiv c \pmod{m}$

Ex 6 :- $R = \{(a, b) \mid a \text{ divides } b\} \quad A = \mathbb{Z}$

Reflexive :- $\forall a \in A \quad (a, a) \in R$.
 $\forall a \in \mathbb{Z} \quad a \text{ divides } a \quad \checkmark$

Symmetric $\forall a, b \in A \quad \text{if } (a, b) \in R \rightarrow (b, a) \in R$.
 $\forall a, b \in \mathbb{Z} \quad \text{if } a \text{ divides } b \rightarrow b \text{ divides } a$
 $\times$

This is not Equivalence Relation.

Ex 7 :- $R = \{(a, b) \mid |a - b| < 1\} \quad A = \mathbb{R}$

Reflexive :- $\forall a \in A \quad (a, a) \in R$.
 $\forall a \in \mathbb{R} \quad |a - a| < 1 \quad \checkmark$

Symmetric $\forall a, b \in A \quad \text{if } (a, b) \in R \rightarrow (b, a) \in R$.
 $\forall a, b \in \mathbb{R} \quad \text{if } |a - b| < 1 \rightarrow |b - a| < 1$
 $\checkmark$

Transitive $\forall a, b, c \in A \quad \text{if } (a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$.
 $\forall a, b, c \in \mathbb{R} \quad \text{if } |a - b| < 1 \wedge |b - c| < 1 \rightarrow |a - c| < 1$.
 $\underset{0.9}{|5 - 4.1|} < 1 \wedge \underset{0.9}{|4.1 - 3.2|} < 1 \rightarrow \underset{1.8 \not< 1}{|5 - 3.2|} < 1$

This is not Equivalence.

Equivalence classes.

Equivalence classes.

$$[a] = \{s \mid (a, s) \in R\}.$$

Ex 8
496

$$R = \{(a, b) \mid a \equiv b \text{ or } a = -b\}. \quad A = \mathbb{Z}$$

$$[7] = \{7, -7\}. \quad \{(7, 7), (7, -7), (-7, 7), (-7, -7)\}.$$

Ex 8
496

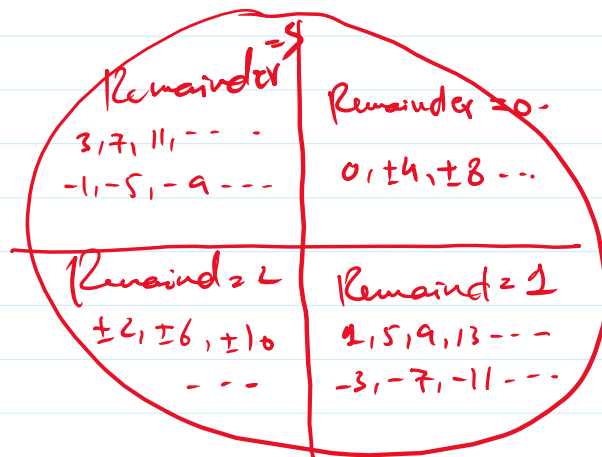
$$R = \{(a, b) \mid a \equiv b \pmod{4}\}. \quad A = \mathbb{Z}.$$

$$[0] = \{0, \pm 4, \pm 8, \pm 12, \dots\}.$$

$$[1] = \{1, 5, 9, 13, \dots, -3, -7, -11, \dots\}.$$

$$[2] = \{2, 6, 10, 14, 18, \dots, -2, -6, -10, \dots\}.$$

$$[3] = \{3, 7, 11, 15, \dots, -1, -5, -9, \dots\}.$$



Partition: $R = \{A_1, A_2, A_3, \dots, A_n\}$ where $\mathcal{P}$ is a family of sets.
is a partition of S if.

$$1) \quad \forall i \quad A_i \neq \emptyset.$$

$$1) \forall i A_i \neq \emptyset.$$

$$2) \forall i, j A_i \cap A_j = \emptyset \quad i \neq j.$$

$$3) \bigcup_{i \in I} A_i = S.$$

Ex 12
498. $S = \{1, 2, 3, 4, 5, 6\}$
 $p = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$ is p a partition.
 $\uparrow A_1 \quad \downarrow A_2 \quad \uparrow A_3$

$$1) \forall i A_i \neq \emptyset = A_1 \neq \emptyset \wedge A_2 \neq \emptyset \wedge A_3 \neq \emptyset.$$

T $\wedge$ T $\wedge$ T $=$ T ✓

$$2) A_1 \cap A_2 = \emptyset \wedge A_2 \cap A_3 = \emptyset \wedge A_1 \cap A_3 = \emptyset.$$

T $\wedge$ T $\wedge$ T $=$ T ✓

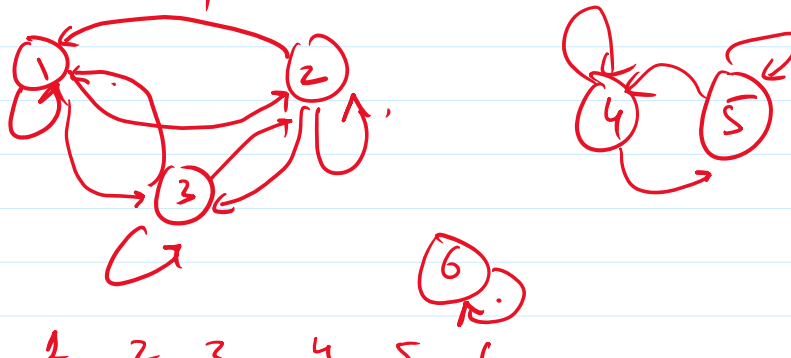
$$3) \bigcup_{i \in I} A_i = S.$$

$$A_1 \cup A_2 \cup A_3 = S.$$

T ✓

Equivalence Relations always creates a partition.

$$p = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}.$$



$$\begin{array}{c}
 \checkmark \qquad \qquad \qquad \textcircled{0} \\
 \qquad \qquad \qquad \nwarrow \\
 \begin{array}{c}
 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \\
 \left[\begin{array}{cccccc}
 1 & 1 & 1 & 0 & 0 & 0 \\
 1 & 1 & 1 & 0 & 0 & 0 \\
 1 & 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 1 & 1 & 0 \\
 0 & 0 & 0 & 1 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1
 \end{array} \right]
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 \end{array}$$